

ELEC 2FH4 Quiz 3

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Question 1 (10 points)

Two concentric conducting cylindrical shells of radii a and b ($b > a$) and length ℓ form a capacitor. The region between the cylinders is filled with a dielectric material of permittivity ϵ . The inner and outer cylinders carry charges $+Q$ and $-Q$, respectively. Assume $\ell \gg b$ so fringing effects can be neglected.

a) Using Gauss's law, determine the electric field intensity $E(\rho)$ in the region

$$a < \rho < b$$

and hence determine the capacitance of the cylindrical capacitor.

b) Solve Laplace's equation in cylindrical coordinates for the potential in the region

$$a < \rho < b: \quad \nabla^2 V = 0$$

Assuming the potential depends only on ρ , apply the boundary conditions to determine the capacitance and show that it matches the result obtained in part (a).

c) Using

$$a = 1 \text{ cm}, \quad b = 3 \text{ cm}, \quad \ell = 30 \text{ cm}, \quad \epsilon = 2.5 \epsilon_0$$

a) $\vec{E} = ? \quad C = ?$

$$Q = \oint \vec{D} \cdot d\vec{A}$$

$$+Q = D(\rho) \cdot 2\pi\rho\ell$$

$$\vec{D}(\rho) = \frac{Q}{2\pi\rho\ell}$$

$$\vec{E}(\rho) = \frac{1}{\epsilon} \vec{D}(\rho)$$

$$= \frac{1}{\epsilon} \frac{Q}{2\pi\rho\ell}$$

$$\vec{E}(\rho) = \frac{Q}{2\pi\epsilon\rho\ell}$$

$$\begin{cases} \vec{D} = \epsilon \vec{E} \\ \vec{E} = \frac{1}{\epsilon} \vec{D} \end{cases}$$

$$\begin{aligned} V_{ab} &= \int_a^b E(\rho) d\rho \\ &= \int_a^b \frac{Q}{2\pi\epsilon\rho\ell} d\rho \\ &= \frac{Q}{2\pi\epsilon\ell} \int_a^b \frac{1}{\rho} d\rho \\ &= \frac{Q}{2\pi\epsilon\ell} \ln\rho \Big|_a^b \\ &= \frac{Q}{2\pi\epsilon\ell} (\ln b - \ln a) \\ &= \frac{Q}{2\pi\epsilon\ell} \ln\left(\frac{b}{a}\right) \end{aligned}$$

$$\begin{aligned} C &= \frac{Q}{V} \\ &= \frac{Q}{V_{ab}} \\ &= \frac{Q}{\frac{Q}{2\pi\epsilon\ell} \ln\left(\frac{b}{a}\right)} \\ C &= \frac{2\pi\epsilon\ell}{\ln\left(\frac{b}{a}\right)} \end{aligned}$$

b) $0 = \nabla^2 V = \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dV}{d\rho} \right)$

$$V = A \ln \rho + B$$

$$V(\rho) = A \ln \rho + B \quad (1)$$

↓

$$V(a) = V_a = V_0: \quad V_0 = A \ln a + B \quad (2)$$

$$V(b) = V_b = 0: \quad 0 = A \ln b + B \quad (3) \rightarrow B = -A \ln b \quad (4)$$

$$V_0 = A \ln a + B \quad (4)$$

$$V_0 = A \ln a - A \ln b$$

$$V_0 = A \ln\left(\frac{a}{b}\right)$$

$$A = V_0 / \ln\left(\frac{a}{b}\right) \quad (5)$$

$$\begin{aligned} (1): \quad V(\rho) &= A \ln \rho + B \\ &= A \ln \rho - A \ln b \\ &= A \ln\left(\frac{\rho}{b}\right) \\ &= V_0 \frac{\ln\left(\frac{\rho}{b}\right)}{\ln\left(\frac{a}{b}\right)} \end{aligned}$$

$$\begin{aligned} E(\rho) &= -\nabla V \\ &= -\frac{dV}{d\rho} \\ &= -\frac{d}{d\rho} \left(V_0 \frac{\ln\left(\frac{\rho}{b}\right)}{\ln\left(\frac{a}{b}\right)} \right) \\ &= -\frac{d}{d\rho} \left(V_0 \cdot \frac{1}{\ln\left(\frac{a}{b}\right)} \cdot \ln\left(\frac{\rho}{b}\right) \right) \\ \vec{E}(\rho) &= V_0 \cdot \frac{1}{\ln\left(\frac{a}{b}\right)} \cdot \frac{1}{\rho} \end{aligned}$$

* from a) $E(\rho) = \frac{Q}{2\pi\epsilon\rho\ell}$

$$\epsilon = \epsilon$$

$$V_0 \cdot \frac{1}{\ln\left(\frac{a}{b}\right)} \cdot \frac{1}{\rho} = \frac{Q}{2\pi\epsilon\rho\ell}$$

$$V_0 \cdot 2\pi\epsilon\ell \cdot \frac{1}{\ln\left(\frac{a}{b}\right)} = Q$$

$$C = \frac{Q}{V_0} = \frac{2\pi\epsilon\ell}{\ln\left(\frac{b}{a}\right)}$$

↳ matches part a)

c) $a = 1 \text{ cm} = 0.01$
 $b = 3 \text{ cm} = 0.03$
 $\ell = 30 \text{ cm} = 0.3$
 $\epsilon = 2.5 \epsilon_0$

$$\begin{aligned} C &= \frac{2\pi\epsilon\ell}{\ln\left(\frac{b}{a}\right)} \\ &= \frac{2\pi \cdot 2.5 \epsilon_0 \cdot (0.3)}{\ln\left(\frac{0.03}{0.01}\right)} \\ &= 3.80 \times 10^{-11} \text{ F} \\ &= 38.0 \times 10^{-12} \text{ F} \\ C &= 38.0 \text{ pF} \end{aligned}$$